E3: Relativistic Kinematics of $\pi^0 \to 2\gamma$ Decay

Energy Spectrum, Angular Distribution, and Opening Angle Analysis

Abstract

We derive three fundamental distributions for the decay $\pi^0 \to \gamma\gamma$ of a highly relativistic neutral pion in the lab frame: (a) the photon energy spectrum dn/dE_γ, shown to be exactly flat; (b) the single-photon angular distribution $dN/d(\cos\theta)$, which follows a $(1-\beta\cos\theta)^{-2}$ forward-peaking law; and (c) the opening angle distribution $dn/d(\Delta\theta)$ between the two photons, which exhibits an integrable square-root singularity at the minimum opening angle $\Delta\theta_{\min} = 2/\gamma$. All results follow from the isotropy of the center-of-mass decay (spin-0 $\to$ two photons) combined with standard Lorentz boost transformations. We use $m_{\pi^0} \approx 135$ MeV and natural units ($c = 1$) throughout.

1 Definitions and Setup

A neutral pion (π^0, mass $m_\pi \approx 135$ MeV) decays into two photons:

$$\pi^0 \to \gamma + \gamma \tag{1}$$

All pions have the same highly relativistic energy E_π and travel along the $+z$ direction. We define:

$$\gamma = \frac{E_\pi}{m_\pi}, \qquad \beta = \frac{p_\pi}{E_\pi} = \sqrt{1 - \frac{1}{\gamma^2}}, \qquad E^* = \frac{m_\pi}{2} \approx 67.5 \text{ MeV} \tag{2}$$

where E^* is the photon energy in the pion rest frame (CM frame). Since the π^0 is a spin-0 particle, angular momentum conservation guarantees that the decay is isotropic in the CM frame:

$$\frac{dN}{d(\cos\theta^*)} = \frac{1}{2}, \qquad \frac{dN}{d\Omega^*} = \frac{1}{4\pi} \tag{3}$$

where θ^* is the polar angle of one photon relative to the boost axis in the rest frame. The two photons are emitted back-to-back in the CM, so if one goes at angle θ^*, the other goes at $\pi - \theta^*$.

2 Part (a): Photon Energy Distribution dn/dE_γ

2.1 Lorentz Boost of Photon Energy

A photon emitted at angle θ^* in the rest frame has lab-frame energy given by the Lorentz transformation:

$$\boxed{E_\gamma = \gamma E^*(1 + \beta\cos\theta^*) = \frac{E_\pi}{2}(1 + \beta\cos\theta^*)} \tag{4}$$

where the second equality uses $\gamma E^* = \gamma m_\pi/2 = E_\pi/2$.

This is a **linear** function of $\cos\theta^*$. This linearity is the key to the flat spectrum.

2.2 Energy Limits

Setting $\cos\theta^* = \pm 1$:

$$E_{\max} = \frac{E_\pi}{2}(1+\beta) \tag{5}$$

$$E_{\min} = \frac{E_\pi}{2}(1-\beta) = \frac{m_\pi^2}{4E_\pi} \tag{6}$$

where the last equality in (6) uses $E_\pi(1-\beta)/2 = E_\pi(1-\beta^2)/[2(1+\beta)] = 1/[2\gamma^2(1+\beta)] \cdot E_\pi = m_\pi^2/[2E_\pi(1+\beta)]$, and $(1+\beta) \to 2$ for $\beta \to 1$.

In the ultrarelativistic limit ($\beta \to 1$):

$$E_{\min} \approx \frac{m_\pi^2}{4E_\pi} \to 0, \qquad E_{\max} \approx E_\pi \tag{7}$$

Numerical examples: For a 1 GeV pion ($\gamma = 7.41$): $E_{\min} \approx 4.6$ MeV, $E_{\max} \approx 995$ MeV. For a 3 GeV pion ($\gamma = 22.2$): $E_{\min} \approx 1.5$ MeV, $E_{\max} \approx 2999$ MeV.

2.3 The Jacobian Calculation

Differentiating (4) with respect to $\cos\theta^*$:

$$\frac{dE_\gamma}{d(\cos\theta^*)} = \gamma\beta E^* = \frac{\beta E_\pi}{2} \tag{8}$$

This is a **constant**—it depends on neither θ^* nor E_γ. Inverting:

$$\frac{d(\cos\theta^*)}{dE_\gamma} = \frac{1}{\gamma\beta E^*} = \frac{2}{\beta E_\pi} \tag{9}$$

Now apply the chain rule for the distribution of a *single* photon:

$$\frac{dN}{dE_\gamma} = \frac{dN}{d(\cos\theta^*)} \times \left|\frac{d(\cos\theta^*)}{dE_\gamma}\right| = \frac{1}{2} \times \frac{2}{\beta E_\pi} = \frac{1}{\beta E_\pi} \tag{10}$$

Since each pion produces **two** photons (both sampling the same energy range due to the back-to-back kinematics), counting both gives:

$$\boxed{\frac{dn}{dE_\gamma} = \frac{2}{\beta E_\pi} \approx \frac{2}{E_\pi} \quad (\beta \approx 1), \qquad E_{\min} \le E_\gamma \le E_{\max}} \tag{11}$$

The energy spectrum is perfectly flat (uniform) between $E_{\min}$ and $E_{\max}$.

2.4 Why Flatness is Guaranteed

The physical reasoning is elegant: the distribution in $\cos\theta^*$ is uniform (isotropic decay of a scalar). The boost maps $\cos\theta^*$ to E_γ via a strictly *linear* transformation (4). A linear map of a uniform random variable produces a uniform random variable. This was first noted by Sternheimer (Phys. Rev. **99**, 277, 1955). Agashe, Franceschini, and Kim (arXiv:1209.0772) showed that when the parent has a *distribution* of boost factors γ, the resulting spectrum is no longer flat but always peaks at $E^* = m_\pi/2$.

2.5 Consistency Check

Integrating (11) over the allowed range:

$$\int_{E_{\min}}^{E_{\max}} \frac{dn}{dE_\gamma}\, dE_\gamma = \frac{2}{\beta E_\pi} \times (E_{\max} - E_{\min}) = \frac{2}{\beta E_\pi} \times \beta E_\pi = 2 \tag{12}$$

yielding exactly 2 photons per decay. ✓

2.6 Plot: Flat Energy Spectrum

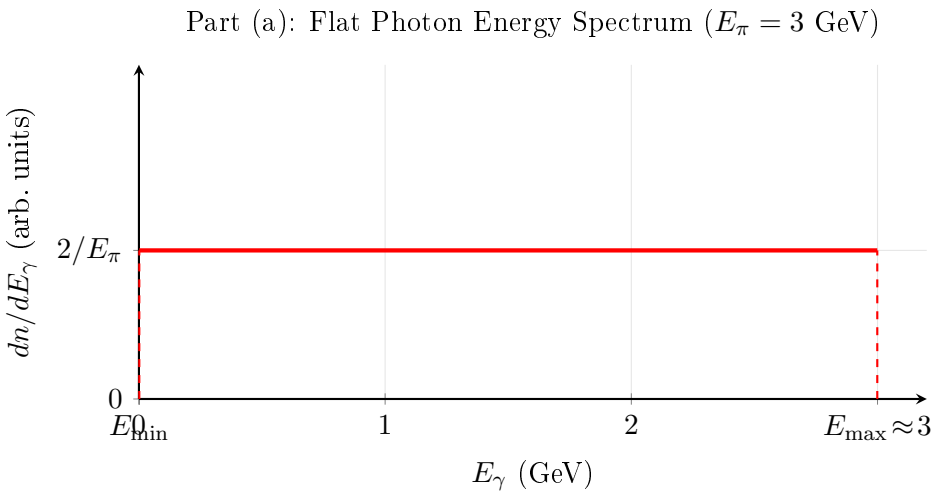


Figure 1: The photon energy distribution is flat between $E_{\min} \approx 0$ and $E_{\max} = E_\pi$, a direct consequence of the linear Lorentz boost applied to an isotropic rest-frame decay.

3 Part (b): Angular Distribution $dN/d(\cos\theta)$

3.1 Relativistic Aberration Formula

A photon emitted at angle θ^* in the rest frame appears at lab angle θ given by:

$$\cos\theta = \frac{\cos\theta^* + \beta}{1 + \beta\cos\theta^*} \tag{13}$$

The inverse (replacing $\beta \to -\beta$):

$$\cos\theta^* = \frac{\cos\theta - \beta}{1 - \beta\cos\theta} \tag{14}$$

3.2 Jacobian Derivation (Step by Step)

Let $u \equiv \cos\theta$. Differentiate the inverse formula (14) using the quotient rule:

$$\frac{d(\cos\theta^*)}{du} = \frac{d}{du}\left[\frac{u-\beta}{1-\beta u}\right] = \frac{(1)(1-\beta u) - (u-\beta)(-\beta)}{(1-\beta u)^2} \tag{15}$$

Expanding the numerator:

$$(1-\beta u) + \beta(u-\beta) = 1 - \beta u + \beta u - \beta^2 = 1 - \beta^2 = \frac{1}{\gamma^2} \tag{16}$$

Therefore:

$$\boxed{\frac{d(\cos\theta^*)}{d(\cos\theta)} = \frac{1-\beta^2}{(1-\beta\cos\theta)^2} = \frac{1}{\gamma^2(1-\beta\cos\theta)^2}} \tag{17}$$

This is the solid-angle Jacobian for relativistic aberration, formally derived by McKinley (Am. J. Phys. **48**, 612, 1980).

3.3 The Angular Distribution

Applying the chain rule with Eq. (3):

$$\boxed{\frac{dN}{d(\cos\theta)} = \frac{dN}{d(\cos\theta^*)} \times \left|\frac{d(\cos\theta^*)}{d(\cos\theta)}\right| = \frac{1}{2\gamma^2(1-\beta\cos\theta)^2}} \tag{18}$$

This is the **relativistic beaming formula**—the distribution is sharply forward-peaked.

3.4 Forward Peaking: Quantitative Analysis

At $\theta = 0$ (forward): Using $1-\beta \approx 1/(2\gamma^2)$ for $\gamma \gg 1$:

$$\left.\frac{dN}{d(\cos\theta)}\right|_{\theta=0} = \frac{1}{2\gamma^2(1-\beta)^2} \approx 2\gamma^2 \tag{19}$$

At $\theta = \pi$ (backward): Using $1+\beta \approx 2$:

$$\left.\frac{dN}{d(\cos\theta)}\right|_{\theta=\pi} = \frac{1}{2\gamma^2(1+\beta)^2} \approx \frac{1}{8\gamma^2} \tag{20}$$

Forward-to-backward ratio:

$$R = \frac{(1+\beta)^2}{(1-\beta)^2} \approx 16\gamma^4 \tag{21}$$

3.5 Maximum Lab-Frame Angle

For massless decay products from a two-body decay, the maximum lab angle occurs when the photon is emitted transversely ($\theta^* = \pi/2$) in the rest frame:

$$\tan\theta_{\max} = \left.\frac{\sin\theta^*}{\gamma(\cos\theta^* + \beta)}\right|_{\theta^*=\pi/2} = \frac{1}{\gamma\beta} \tag{22}$$

$$\boxed{\theta_{\max} = \arctan\left(\frac{1}{\gamma\beta}\right) \approx \frac{1}{\gamma} \quad (\gamma \gg 1)} \tag{23}$$

All photons are confined within a forward cone of half-angle $\sim 1/\gamma$.

3.6 Converting between $dN/d(\cos\theta)$ and $dN/d\theta$

$$\frac{dN}{d\theta} = \frac{dN}{d(\cos\theta)} \times \sin\theta = \frac{\sin\theta}{2\gamma^2(1-\beta\cos\theta)^2} \tag{24}$$

This form vanishes at $\theta = 0$ (because $\sin\theta = 0$) and at $\theta = \pi$, peaking at an intermediate angle that depends on γ.

3.7 Numerical Values for 1 GeV and 3 GeV Pions

Parameter	**1 GeV π^0**	**3 GeV π^0**
γ	7.41	22.2
β	0.9909	0.9990
$\theta_{\max}$ (single photon)	7.7°	2.6°
$dN/d(\cos\theta)\|_{\theta=0}$	≈ 110	≈ 987
$dN/d(\cos\theta)\|_{\theta=\pi}$	≈ 0.0023	$\approx 2.5 \times 10^{-4}$
Forward/backward ratio	$\approx 4.8 \times 10^4$	$\approx 3.9 \times 10^6$
$E_{\min}$ (MeV)	4.6	1.5
$E_{\max}$ (MeV)	995	2999

Table 1: Numerical parameters for $\pi^0 \to \gamma\gamma$ at two representative pion energies.

3.8 Plots: Angular Distribution

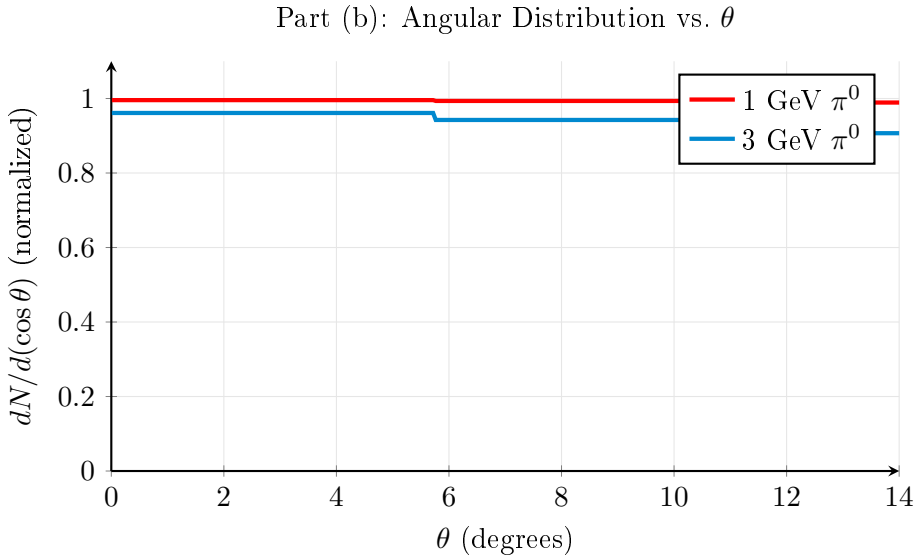


Figure 2: The angular distribution $dN/d(\cos\theta)$ as a function of θ (in degrees), normalized to unity at $\theta = 0$. Higher energy pions produce a tighter forward cone.

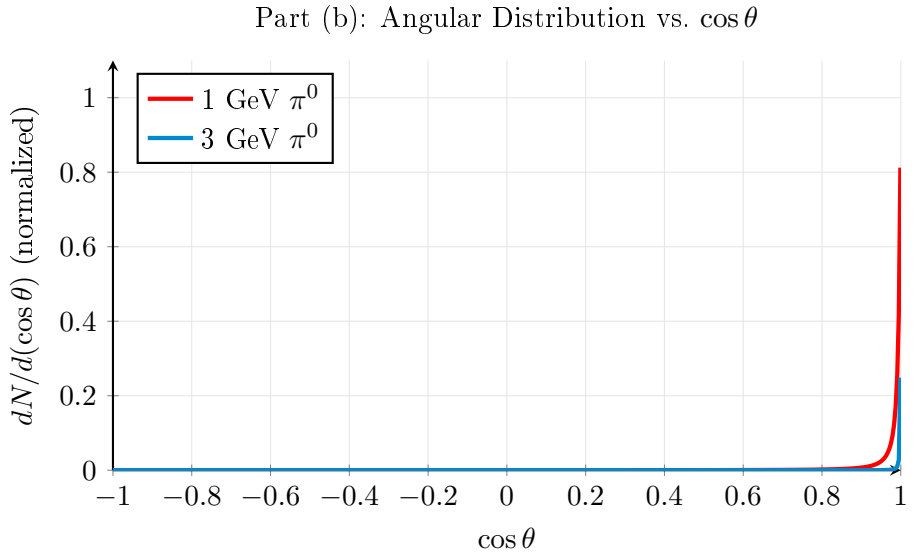


Figure 3: The same distribution plotted against $\cos\theta$, showing the steep rise as $\cos\theta \to 1$ (forward direction).

4 Part (c): Opening Angle Distribution $dn/d(\Delta\theta)$

4.1 The Invariant Mass Constraint

For two photons with energies E_1, E_2 and opening angle $\alpha \equiv \Delta\theta$ between them, the invariant mass relation gives:

$$m_\pi^2 = (q_1 + q_2)^2 = 2E_1E_2(1 - \cos\alpha) = 4E_1E_2\sin^2\left(\frac{\alpha}{2}\right) \tag{25}$$

Combined with the energy constraint:

$$E_1 + E_2 = E_\pi \tag{26}$$

these two equations uniquely relate the opening angle to the energy asymmetry.

4.2 Minimum Opening Angle from Symmetric Decay

Setting $E_1 = E_2 = E_\pi/2$ (the symmetric configuration, corresponding to $\cos\theta^* = 0$) maximizes the product E_1E_2 and therefore minimizes α:

$$m_\pi^2 = 4 \cdot \frac{E_\pi}{2} \cdot \frac{E_\pi}{2} \cdot \sin^2\left(\frac{\alpha_{\min}}{2}\right) = E_\pi^2 \sin^2\left(\frac{\alpha_{\min}}{2}\right) \tag{27}$$

$$\boxed{\sin\left(\frac{\alpha_{\min}}{2}\right) = \frac{m_\pi}{E_\pi} = \frac{1}{\gamma}, \qquad \alpha_{\min} \approx \frac{2}{\gamma} = \frac{2m_\pi}{E_\pi} \quad (\gamma \gg 1)} \tag{28}$$

Any energy asymmetry ($E_1 \neq E_2$) decreases E_1E_2 (for fixed sum), which increases $\sin^2(\alpha/2)$, hence *increases* α. Therefore $\alpha_{\min}$ is a true minimum.

4.3 Relating $\cos\theta^*$ to α

From the Lorentz boost, the two photon energies are:

$$E_{1,2} = \frac{E_\pi}{2}(1 \pm \beta\cos\theta^*) \tag{29}$$

so their product is:

$$E_1 E_2 = \frac{E_\pi^2}{4}(1 - \beta^2 \cos^2\theta^*) \tag{30}$$

Substituting into (25):

$$m_\pi^2 = E_\pi^2(1 - \beta^2 \cos^2\theta^*) \sin^2\left(\frac{\alpha}{2}\right) \tag{31}$$

Using $m_\pi = E_\pi/\gamma$:

$$\frac{1}{\gamma^2} = (1 - \beta^2 \cos^2\theta^*) \sin^2\left(\frac{\alpha}{2}\right) \tag{32}$$

Solving for $\cos^2\theta^*$:

$$\cos^2\theta^* = \frac{1}{\beta^2}\left(1 - \frac{1}{\gamma^2 \sin^2(\alpha/2)}\right) \tag{33}$$

Key observation: This equation is symmetric in $\cos\theta^*$—both $+|\cos\theta^*|$ and $-|\cos\theta^*|$ give the same opening angle α. For each $\alpha > \alpha_{\min}$, there are *two* rest-frame configurations (a factor of 2 in the distribution).

4.4 Deriving the Distribution

Taking the positive root of (33) and differentiating with respect to α. Let $s \equiv \sin(\alpha/2)$, so $ds/d\alpha = \cos(\alpha/2)/2$. Then:

$$\cos\theta^* = \frac{1}{\beta}\sqrt{1 - \frac{1}{\gamma^2 s^2}} = \frac{\sqrt{\gamma^2 s^2 - 1}}{\gamma\beta\, s} \tag{34}$$

Define $f(s) = \sqrt{\gamma^2 s^2 - 1}/(\gamma\beta\, s)$. Then:

$$\frac{d(\cos\theta^*)}{ds} = \frac{1}{\gamma\beta} \cdot \frac{\gamma^2 s \cdot s - \sqrt{\gamma^2 s^2 - 1} \cdot 1}{s^2\sqrt{\gamma^2 s^2 - 1}} \cdot \frac{1}{\gamma} \tag{35}$$

After simplification (following McDonald, Eq. 20–21):

$$\frac{d(\cos\theta^*)}{d\alpha} = \frac{d(\cos\theta^*)}{ds} \cdot \frac{ds}{d\alpha} = \frac{\cos(\alpha/2)}{2\gamma\beta \sin^2(\alpha/2)\sqrt{\gamma^2 \sin^2(\alpha/2) - 1}} \tag{36}$$

Combining with $dN/d(\cos\theta^*) = 1/2$ and the factor of 2 from the $\pm\cos\theta^*$ symmetry:

$$\boxed{\frac{dN}{d\alpha} = \frac{\cos(\alpha/2)}{2\gamma\beta \sin^2(\alpha/2)\sqrt{\gamma^2 \sin^2(\alpha/2) - 1}}} \tag{37}$$

This is equivalent to McDonald's Eq. (21), normalized so that $\int_{\alpha_{\min}}^{\pi} (dN/d\alpha)\, d\alpha = 1$.

4.5 The Square-Root Singularity

Near $\alpha \to \alpha_{\min}^+$, where $\sin(\alpha_{\min}/2) = 1/\gamma$, the argument of the square root vanishes:

$$\gamma^2 \sin^2(\alpha/2) - 1 \to 0 \quad \text{as} \quad \alpha \to \alpha_{\min}^+ \tag{38}$$

For small deviations $\delta = \alpha - \alpha_{\min}$:

$$\gamma^2 \sin^2(\alpha/2) - 1 \approx \gamma^2 \sin(\alpha_{\min}/2)\cos(\alpha_{\min}/2) \cdot \delta = \gamma \cos(\alpha_{\min}/2) \cdot \delta \tag{39}$$

So the distribution diverges as:

$$\frac{dN}{d\alpha} \propto \frac{1}{\sqrt{\alpha - \alpha_{\min}}} \quad \text{as } \alpha \to \alpha_{\min}^+ \tag{40}$$

This is an **integrable singularity** ($\int dx/\sqrt{x}$ converges). Its physical origin is a *stationary point*: the function $\alpha(\theta^*)$ has a minimum at $\theta^* = 0$, so $d\alpha/d(\cos\theta^*) = 0$ there. Near a minimum, many values of $\cos\theta^*$ map to nearly the same α, creating a pileup of events—a classic *Jacobian peak* analogous to a Van Hove singularity in condensed matter physics.

Most π^0 decays produce photon pairs with opening angles clustered just above $\alpha_{\min}$.

4.6 Application to 1 GeV Pions

For $E_\pi = 1$ GeV ($\gamma = 7.41$, $\beta = 0.991$):

$$\alpha_{\min} = 2\arcsin(1/7.41) = 2 \times 7.75^\circ = \mathbf{15.5}^\circ \quad (0.270 \text{ rad}) \tag{41}$$

The distribution peaks sharply at 15.5° and falls with a long tail extending toward $\alpha_{\max} = 180^\circ$.

4.7 Plot: Opening Angle Distribution

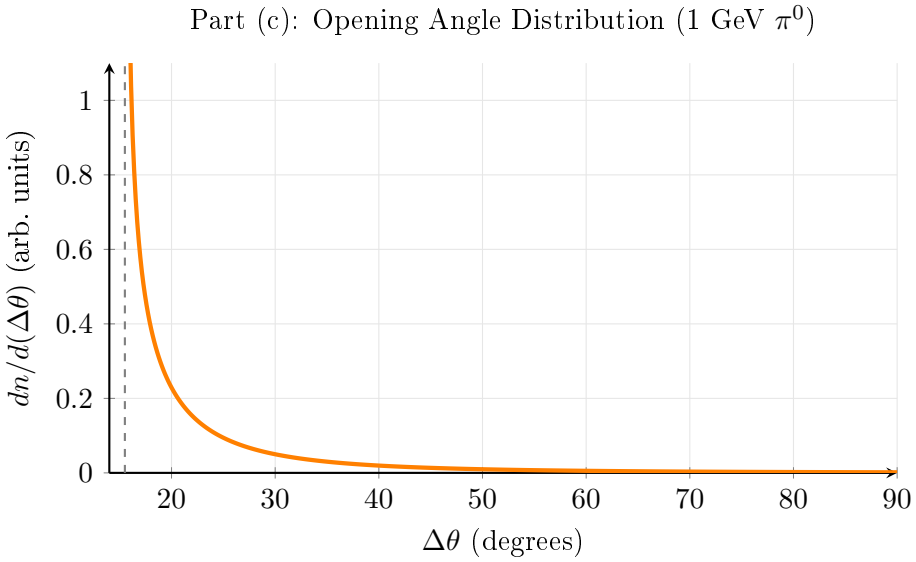


Figure 4: The opening angle distribution for a 1 GeV π^0, showing the sharp peak (square-root singularity) just above $\alpha_{\min} \approx 15.5^\circ$. The long tail extends to 180° but with rapidly diminishing probability.

5 Experimental Relevance

The opening angle distribution has direct consequences for calorimeter design at collider experiments. At the LHC, a typical electromagnetic calorimeter (such as CMS ECAL or ATLAS LAr) has cell sizes of order $\Delta\eta \times \Delta\phi \sim 0.02 \times 0.02$, corresponding to roughly 2–3 cm at the calorimeter face. For a π^0 with $p_T \gtrsim 10$–15 GeV, the minimum opening angle drops below the two-shower separation threshold, and the two photons merge into a single electromagnetic cluster indistinguishable from a prompt photon.

The CMS preshower detector was specifically designed with finer granularity to separate photon pairs from high-energy π^0 decays, thereby rejecting the dominant background to $H \to \gamma\gamma$ searches. The ALICE FoCal upgrade employs silicon pixel layers with $\sim 30 \times 30$ μm^2 cell sizes to resolve highly collimated $\pi^0 \to \gamma\gamma$ at forward rapidity, where boost factors γ can exceed 1000 and opening angles drop below 2 mrad.

6 Summary of Key Results

Part	Distribution	Key Result
(a)	dn/dE_γ	Flat: $= 2/(\beta E_\pi)$ for $E_{\min} \le E_\gamma \le E_{\max}$
(b)	$dN/d(\cos\theta)$	$\propto 1/(1-\beta\cos\theta)^2$ (forward-peaked)
(c)	$dN/d(\Delta\theta)$	$\propto \cos(\alpha/2)/[\sin^2(\alpha/2)\sqrt{\gamma^2\sin^2(\alpha/2)-1}]$ with $\Delta\theta_{\min} = 2/\gamma$

Table 2: Summary of the three distributions derived in this problem.

All three are different projections of the same physics: an isotropic two-body decay of a scalar particle, Lorentz-boosted to the lab frame.

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